

RBF-FD Solver: Least Squares Method for Operator Reproduction

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9/2/2026

1 Weight Constraint RBF-FD Setup:

Standard RBF-FD methods assume a native space for the solution set of a PDE problem in form of:

$$-Lu = -\nabla \cdot (A\nabla u) = f \quad \text{in } \Omega, \quad (1)$$

A is an symmetric positive definite anisotropic tensor and f is the forcing term acting in the domain Ω . The problem is subject to Neumann and Dirichlet Boundary Conditions partitioned within $\partial\Omega$ of the form:

$$u = g_D \quad \text{on } \Gamma_D \subset \partial\Omega, \quad (2)$$

$$\frac{\partial u}{\partial n} \equiv n \cdot A\nabla u = g_N \quad \text{on } \Gamma_N \subset \partial\Omega, \quad (3)$$

Rather than assuming a form for the solution, our goal is to approximate the operator through weights dual to the u sampled at nodes. We define the radial basis functions in the form $\phi_i(\mathbf{x}) := \phi(\|\mathbf{x} - \mathbf{y}_i\|)$ for $i = 1, \dots, m$ and centers $\{\mathbf{y}_i\}_{i=1}^m \subseteq \Omega$. Define the vector space $V_Y = \text{span}\{\phi_1, \dots, \phi_m\}$. Let $X = \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subseteq \Omega$ be a set of stencil nodes in Ω . For the following procedure, assume $m > n$, thus more center nodes than stencil nodes per stencil. We can now define $\Phi \in \mathbb{R}^{n \times m}$, where $\Phi_{ij} = \phi_j(\mathbf{x}_i)$. The role of these radial basis functions will be to approximate the operator L . Rather than assuming exactness of the form $L\phi_j(\mathbf{x}) = \sum_i^n w_i(\mathbf{x})\phi_j(\mathbf{x}_i)$ for $j = 1, \dots, m$ as in the classical RBF-FD, we define a residual function:

$$r_j(\mathbf{x}) := L\phi_j(\mathbf{x}) - \sum_{i=1}^n w_i(\mathbf{x})\phi_j(\mathbf{x}_i) \quad (4)$$

$$r(\mathbf{x}) = L\phi(\mathbf{x}) - \Phi^T \mathbf{w}(\mathbf{x}), \quad \mathbf{w} : \mathbb{R}^n \rightarrow \mathbb{R}^n \text{ for } \mathbf{x} \in \Omega. \quad (5)$$

Suppose $\{p_i\}_{i=1}^l$ is a set of polynomials satisfying the nullspace condition for the radial basis functions. Let $\Pi_k = \text{span}\{p_i\}_{i=1}^l$ with polynomial degree up to k . We want to reproduce the operator exactly on the polynomial basis:

$$Lp_j(\mathbf{x}) := \sum_{i=1}^n w_i(\mathbf{x})p_j(\mathbf{x}_i) \text{ for } j = 1, \dots, l \quad (6)$$

$$Lp(\mathbf{x}) = \mathbf{P}^T \mathbf{w} \quad (7)$$

where $\mathbf{P}_{ij} := p_j(\mathbf{x}_i) \in \mathbb{R}^{n \times l}$. Thus, we have a constrained least squares problem in the form:

$$\begin{cases} \min J(\mathbf{w}(\mathbf{x})) := \min \frac{1}{2} \|\mathbf{r}(\mathbf{x})\|_2^2 & \text{for } \mathbf{x} \in \Omega \\ \text{subject to } \Phi^T \mathbf{w} = Lp \end{cases} \quad (8)$$

We can solve this setup by defining the Lagrangian $\mathcal{L}$ for the problem as follows:

$$\begin{aligned} \mathcal{L}(\mathbf{w}, \boldsymbol{\nu}) &:= J(\mathbf{w}) + \langle \boldsymbol{\nu}, \mathbf{P}^T \mathbf{w} - Lp(\mathbf{x}) \rangle \\ &= \frac{1}{2} \langle \Phi^T \mathbf{w}, \Phi^T \mathbf{w} \rangle - \langle \Phi^T \mathbf{w}, L\phi \rangle + \frac{1}{2} \langle L\phi, L\phi \rangle + \langle \boldsymbol{\nu}, \mathbf{P}^T \mathbf{w} - Lp \rangle \end{aligned}$$

2 Consistency Conditions for Finite Spaces and Interpolation Results

The resulting KKT equations for this system are then:

$$\nabla_w \mathcal{L} = 0 : \Phi L \phi = \Phi \Phi^T w + P \nu \quad (9)$$

$$\nabla_\nu \mathcal{L} = 0 : P^T w = L p \quad (10)$$

The augmented matrix then takes the form:

$$\begin{bmatrix} \Phi \Phi^T & P \\ P^T & 0 \end{bmatrix} \begin{bmatrix} w \\ \nu \end{bmatrix} = \begin{bmatrix} \Phi L \phi \\ L p \end{bmatrix} \quad (11)$$

The primary goal is to approximate $Lu(x) \approx w^T(x)u$, with $u = [u(x_1), \dots, u(x_n)]^T$. Define $L_h f(x) := w^T(x)f$ for some function f . Thus, we have the following:

$$\begin{aligned} L_h f(x) &= w^T(x)f \\ &= [w^T \quad \nu^T] \begin{bmatrix} f \\ 0 \end{bmatrix} \\ &= \left(\begin{bmatrix} \Phi \Phi^T & P \\ P^T & 0 \end{bmatrix}^{-1} \begin{bmatrix} \Phi L \phi \\ L p \end{bmatrix} \right)^T \begin{bmatrix} f \\ 0 \end{bmatrix} \\ &= [L \phi^T \Phi^T \quad L p^T] \begin{bmatrix} \Phi \Phi^T & P \\ P^T & 0 \end{bmatrix}^{-T} \begin{bmatrix} f \\ 0 \end{bmatrix} \\ &= [L \phi^T \Phi^T \quad L p^T] \begin{bmatrix} q \\ \alpha \end{bmatrix} \\ &= L \phi^T(x) (\Phi^T q) + L p^T(x) \alpha \end{aligned}$$

Next, we can show that the coefficients q, α relate to the coefficients of u as follows:

$$\begin{bmatrix} f \\ 0 \end{bmatrix} = \begin{bmatrix} \Phi \Phi^T & P \\ P^T & 0 \end{bmatrix} \begin{bmatrix} q \\ \alpha \end{bmatrix} \quad (12)$$

Define $V_h = \{\rho(x) : \rho(x) = \sum_{i=1}^m \lambda_i \phi_i(x) \text{ such that } \lambda \in \text{Range}(\Phi^T)\} \subseteq V_Y$. Let $s \in V_h + \Pi_k$. Thus, $s(x) = \phi^T(x) \Phi^T q + P \alpha$ where $q \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}^l$. Then:

$$s = s(X) = \Phi \Phi^T q + P \alpha = f(X) = f \quad (13)$$

$$P^T q = 0 \quad (14)$$

Hence, we get interpolation of f at the node set X . Furthermore, from the KKT matrix we can show that $\Phi r = P \nu$ which implies that $\Phi r \in \text{range}(P) = (\ker(P^T))^\perp$. Thus, we get the relationship:

$$r^T (\Phi^T z) = 0 \text{ for } P^T z = 0 \quad (15)$$

Finally, we check the consistency error for $s \in V_h + \Pi_k$ with $\mathbf{P}^T \mathbf{q} = \mathbf{0}$.

$$\begin{aligned}
Ls(x) - L_h s(x) &= L(\phi^T(x) \Phi^T \mathbf{q} + \mathbf{p}^T(x) \alpha) - \mathbf{w}^T(x) s \\
&= L\phi^T(x) \Phi^T \mathbf{q} + L\mathbf{p}^T(x) \alpha - \mathbf{w}^T(x) (\Phi \Phi^T \mathbf{q} + \mathbf{P} \alpha) \\
&= L\phi^T(x) \Phi^T \mathbf{q} + L\mathbf{p}^T(x) \alpha - \mathbf{w}^T(x) \Phi \Phi^T \mathbf{q} - \mathbf{w}^T(x) \mathbf{P} \alpha \\
&= (L\phi(x) - \Phi^T \mathbf{w}(x))^T \Phi^T \mathbf{q} + (L\mathbf{p}^T(x) - \mathbf{P}^T \mathbf{w}(x))^T \alpha \\
&= \mathbf{r}^T \Phi^T \mathbf{q} + (L\mathbf{p}(x) - \mathbf{P}^T \mathbf{w}(x))^T \alpha
\end{aligned}$$

$$Ls(x) - L_h s(x) = 0 \text{ due to (10) and (15)} \quad (16)$$

3 Consistency Results for Function Approximations

Let u be a solution to (1) and $s \in V_h + \Pi_k$. We can say $u = s + (u - s)$ to get the following:

$$\begin{aligned}
\mathcal{E}[u](\mathbf{x}) &:= Lu(\mathbf{x}) - L_h u(\mathbf{x}) \\
&= Lu(\mathbf{x}) - \mathbf{w}^T(\mathbf{x}) \mathbf{u} \\
&= L((u - s) + s)(\mathbf{x}) - \mathbf{w}^T(\mathbf{x}) \mathbf{u} - L_h s(\mathbf{x}) + L_h s(\mathbf{x}) \\
&= L(u - s)(\mathbf{x}) + (Ls(\mathbf{x}) - L_h s(\mathbf{x})) + L_h(s - u)(\mathbf{x}) \\
&= L(u - s)(\mathbf{x}) - \mathbf{w}^T(\mathbf{x}) (\mathbf{u} - \mathbf{s}), \text{ by (16)} \\
&= Le(\mathbf{x}) - \mathbf{w}^T(\mathbf{x}) \mathbf{e}, \mathbf{e}(\mathbf{x}) = u(\mathbf{x}) - s(\mathbf{x}) \\
&= \mathcal{E}[\mathbf{e}](\mathbf{x})
\end{aligned}$$